Large-Scale and Multi-Structured Databases

Project Design

GameZone

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CROSSLAB Innovation for industry 4.0

Application Highlights

Distributed Video Games Social Media

Game Discovery

- Browse trending, top-rated and recommended games
- Search by title, genre, category, OS etc.
- Order search results by name, rating, price and release date
- See reviews, positive rating percentage and games' popularity

Users' Experience

- Add games to the “playing” and “played” list
- Submit reviews (like/dislike + description) for played games
- Send friend requests, accept friendships and explore friends' profiles
- Get personalized game recommendations based on your preferences

Administrative Operations

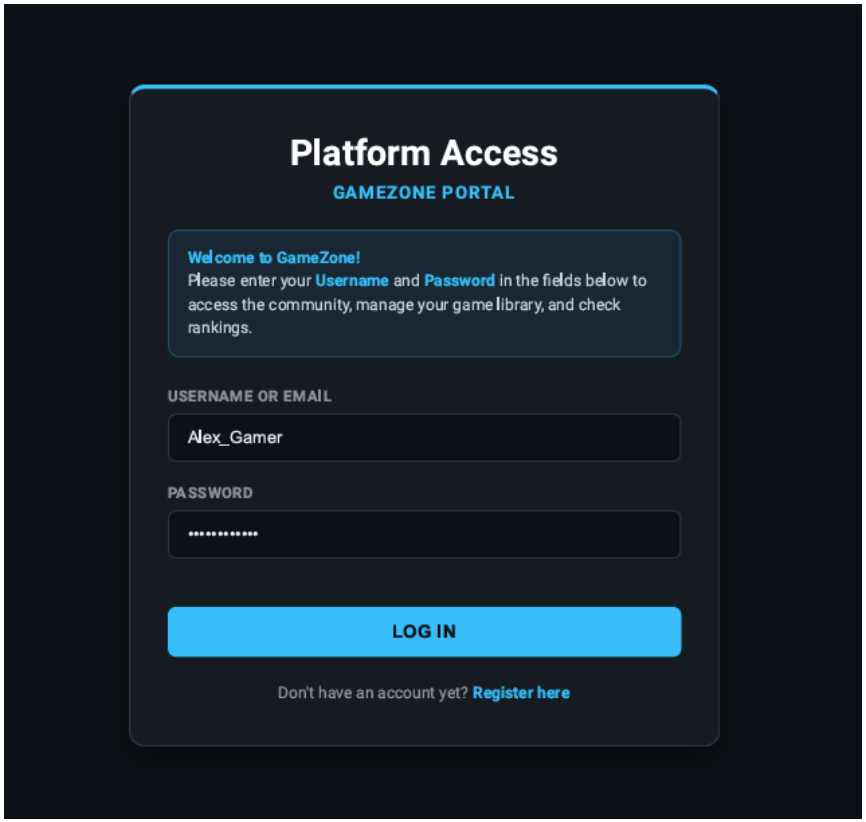
- Moderate users' behaviors (delete inappropriate reviews, ban or delete users)
- Add and remove games, genres and categories from the catalogue

Actors and main mock-ups (i)

Actors:

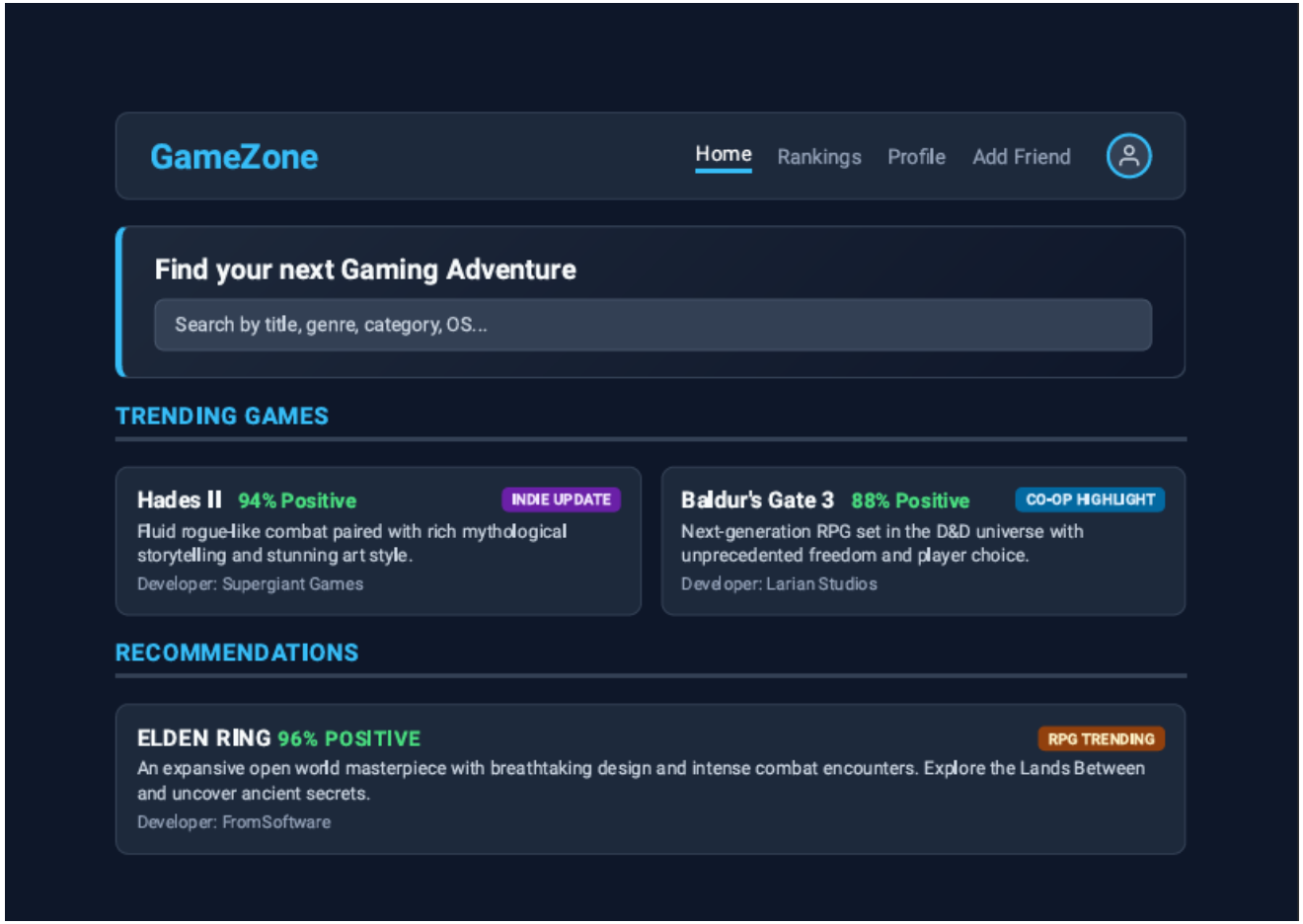
- Unregistred User
- Registered User
- Admin

Login Page



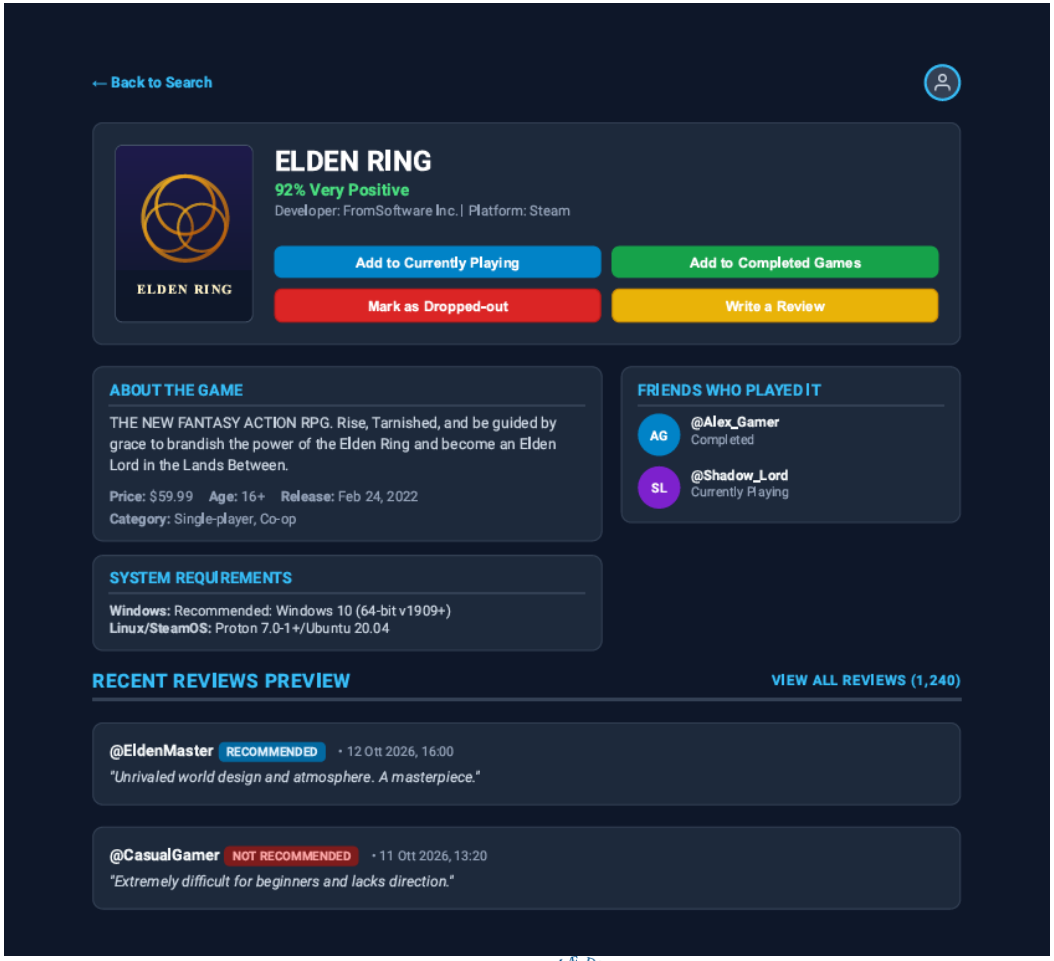
Actors and main mock-ups (ii)

User Browsing Page



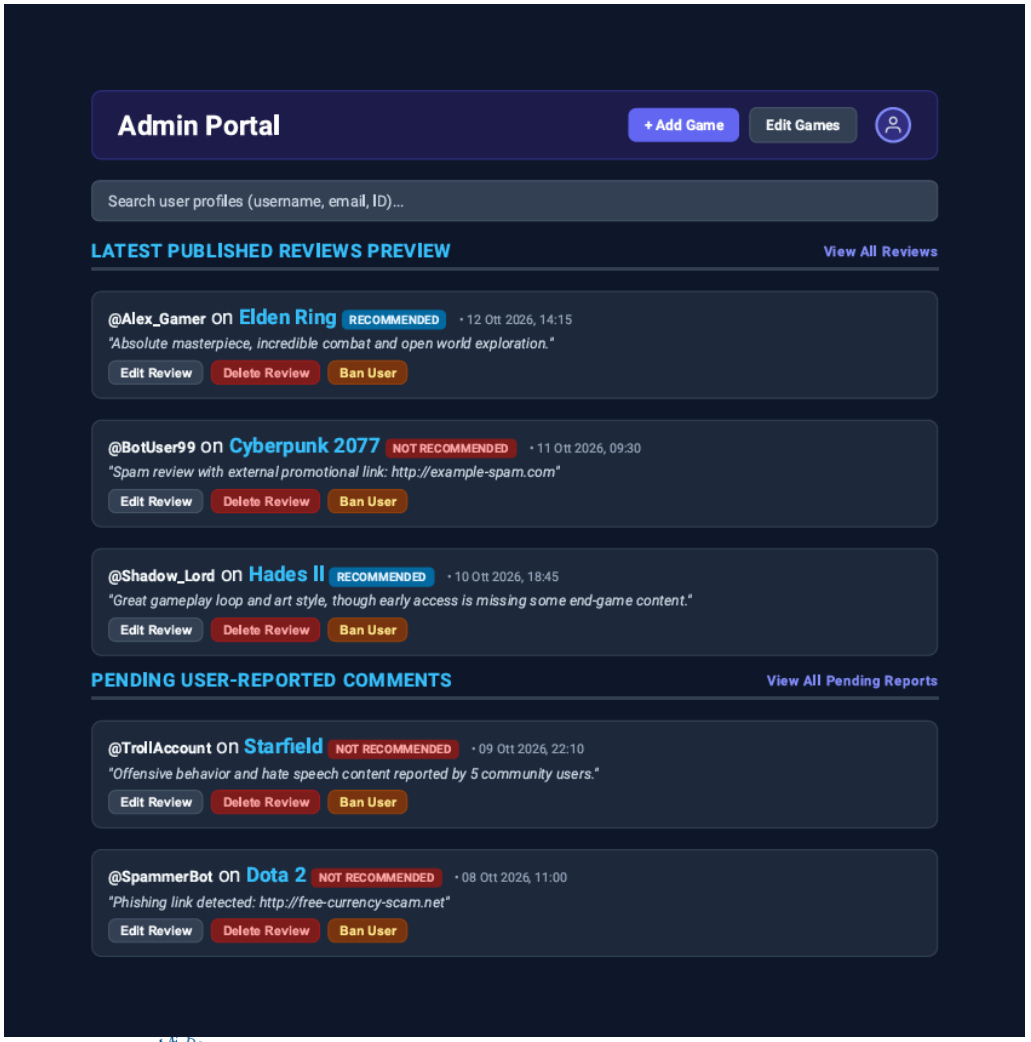
Actors and main mock-ups (iii)

Game Discovery



Actors and main mock-ups (iii)

Admin Page



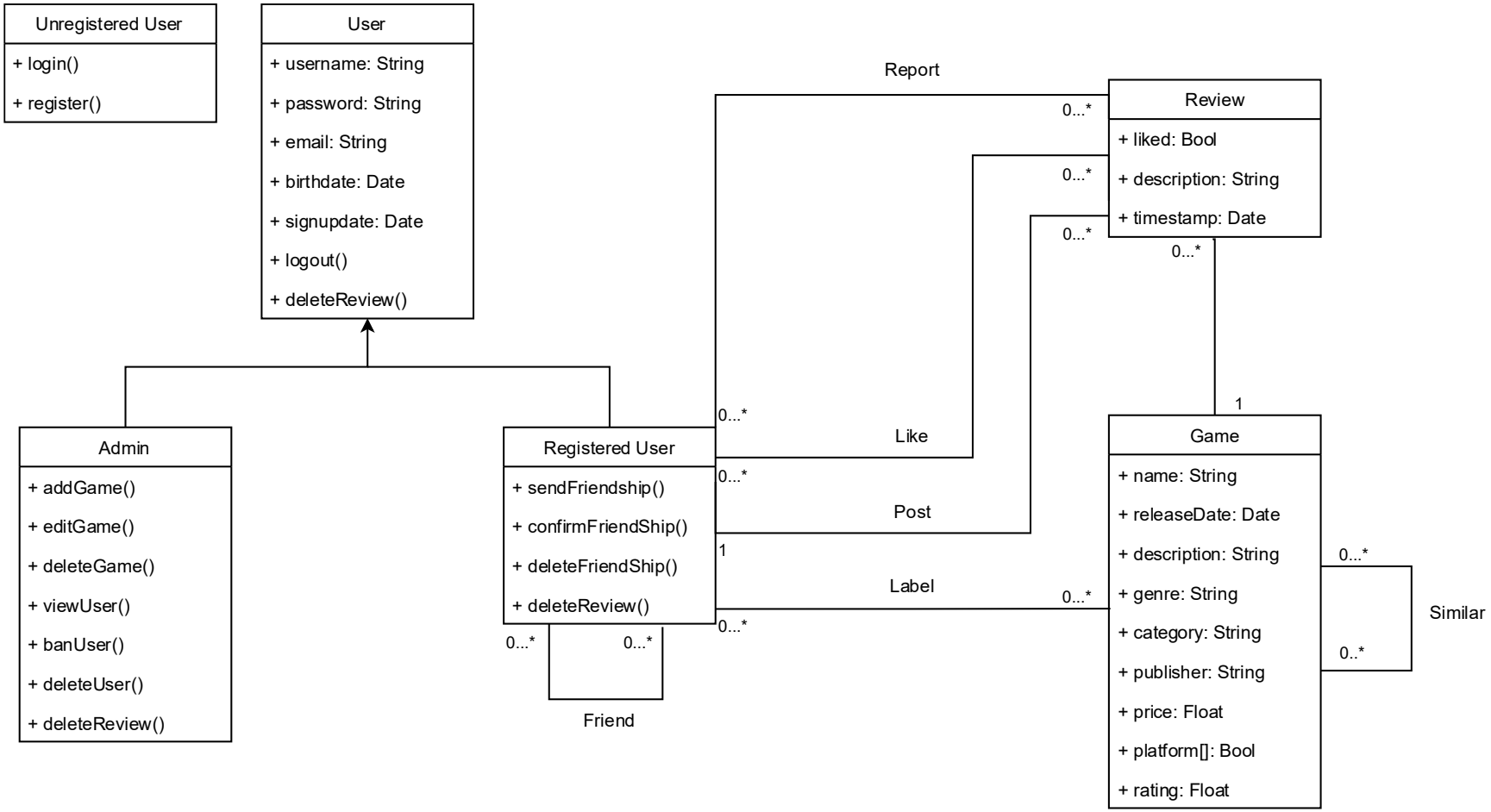
Dataset Description

Sources (from kaggle.com)	
Steam Store Games by *Nik Davis* Information on videogames including title, description, release date, publisher, pricing, OS support, categories and genres.	**Steam Reviews** by *Larxel* Videogame reviews featuring a textual feedback and a binary review.
Original volume: 252.04 MB **Preprocessed**: 36.1 MB (12.814 games, 70% of them have reviews in the preprocessed Steam Reviews)	**Original volume**: 2.16 GB **Preprocessed**: 194 MB (368.961 reviews)

Variety: The preprocessed dataset provides structured data (such as title, appid, price, release date), semi-structured data (categories, genres, requirements) and unstructured data (textual feedbacks, games descriptions).

Velocity/Variability: The preprocessed dataset includes both rapidly changing data (e.g., reviews) and slowly changing information (e.g., game information).

Preliminary UML Class Diagram



Relevant Queries and Entities handled by Document DB

List the most relevant functional requirements supported by the Document DB

- The system must compute ratings for games
- The system must find the highest rated games by genre/category/publisher
- The system must allow admin to ban/delete users
- The system must find the current trending games

List which are the entities stored in the Document DB

- Game
- User
- Review

Requirements and Entities handled by Graph DB

List the most relevant functional requirements supported by the Graph DB

- The system must recommend games similar to those played by the user
- The system must allow the creation of friendships between users
- The system must allow a user to view another user's profile
- The system must display the friends who have played a specified game
- The system must recommend games played by users similar to the current user

List which are the entities stored by the Graph DB

- Game
- User
- Review

